

Final TP 12-14: Putting everything together

SCT – Services and Cloud Technologies

M1 CR/CNI

The objective of these last few TP sessions is to apply everything you have learnt about cloud services to deploy a new application and to make this deployment as “professional” as possible (given the time constraints, of course). For this lab you will have a lot of freedom: you will be requested to choose which cloud technology is the most suitable for your purposes, to make a first deployment of the application, and then to choose which objectives (out of a list of suggestions) you want to target. The goal is to allow you to push things as far as possible, knowing that nobody is expected to complete the full list of objectives in just a few hours of work.

We will be using a well-known open-source micro-service application called Sock Shop. This application is composed of 7 microservices which together constitute a small e-commerce website to sell socks.

- Source code and documentation:

<https://github.com/microservices-demo/microservices-demo>

- Load generator:

<https://github.com/microservices-demo/load-test>

The list of objectives is as follows, together with a rough classification of their level of priority:

- **High:** choose which type of cloud services you want to use (and be ready to argue about your choice). *Please let your tutor know as early as possible which cloud services you will need: AWS, Docker, Kubernetes, Grid’5000, or anything else. Tutors will try their best to give you access to these services (within reasonable limits, of course).*
- **High:** make a basic cloud deployment of the application, at least with some subset of the available services.
- **High:** (when your TP is ready) prepare a screencast or screenshot demo of your working application, and include this demo in your final documentation.
- **Medium:** add support for elasticity, at least for some of the services (you will probably want to focus on the services which need elasticity the most, how will you identify them?)
- **Medium:** add support for automatically generating a workload to stress-test your application.
- **Low:** add support for monitoring the health and performance metrics of your application.
- **Low:** add support for a simple form of automatic autoscaling. How will your system detect that it needs to add/remove resources, and how will you automate the addition/removal operation?
- **Low:** add support for automatic restarting in case some of your microservices crash.

QUESTION: write a report to present the results of your work. Which cloud technologies did you decide to use, and why? Which objectives did you try to reach, and which ones did you actually manage to reach? Which issues did you face and how did you address them? What is the final status of your application?

Your report should in particular include:

- A detailed installation guide to explain how one may reproduce the deployment of your application.
- A demo of your working system. This may be done using screenshots or (even better) a small screencast video whose URL you will give in the report.

— the end —